ATTACHMENT II. SAMOA: DEBT SUSTAINABILITY ANALYSIS[1]

Main message:

- *Samoa continues to be at low risk of external debt distress. Although the debt ratios will rise significantly over the next three to four years as the government receives foreign financing to recover from the recent tsunami, the debt outlook is expected to improve over the medium and long term.*
- *This scenario is predicated on the assumption that the government of Samoa continues to manage its existing debt well and contracts new debt on concessional terms.*
- *Going forward, it would also be important to develop a domestic treasury-bill and bond market.*

Background

1. **Samoa's stock of external debt as of June 2009 was estimated at about US$200 million (or nearly 40 percent of GDP)**. In net-present-value (NPV) terms, the external debt stock was below 30 percent of GDP—a relatively low ratio for a low-income country. External private debt statistics are non available. Newly received partial information suggests that publicly guaranteed debt to SOEs could amount to about 4 percent of GDP. Most of Samoa's public debt is external, with about 85 percent contracted with multilateral creditors on concessional terms, and the remainder with official bilateral creditors.

Samoa: Structure of the External Public Debt, 2009 1/

	In millions of US$	In percent of GDP	NPV in percent of GDP
Public debt	212.8	38.3	28.2
Multilaterals	178.5	32.2	23.2
IMF	0.0	0.0	0.0
World Bank	79.6	14.3	9.3
AsDB	87.1	15.7	12.2
Other multilaterals	11.8	2.1	1.8
Official bilateral	34.3	6.2	5.0
Paris Club	0.7	0.1	0.1
Non-Paris Club	33.6	6.1	4.9
Commercial	0.0	0.0	0.0

Sources: Samoan Ministry of Finance, AsDB, World Bank; and Fund staff estimates.

1/ Fiscal year ending in June.

2. **Samoa has made significant progress in reducing its public debt burden over the last several years.** This improvement reflects prudent fiscal and debt management policies (Attachment Box I), and robust growth. During 2002–08 the external debt indicators improved remarkably: the external debt felt from over 60 percent of GDP to 30 percent of GDP. The ratio of debt service to exports dropped to 4 percent in 2008 from over 10 percent back in 2002.

[1] Prepared by Patrizia Tumbarello (APD).

Attachment Box 1. Samoa: Public Debt Management Strategy

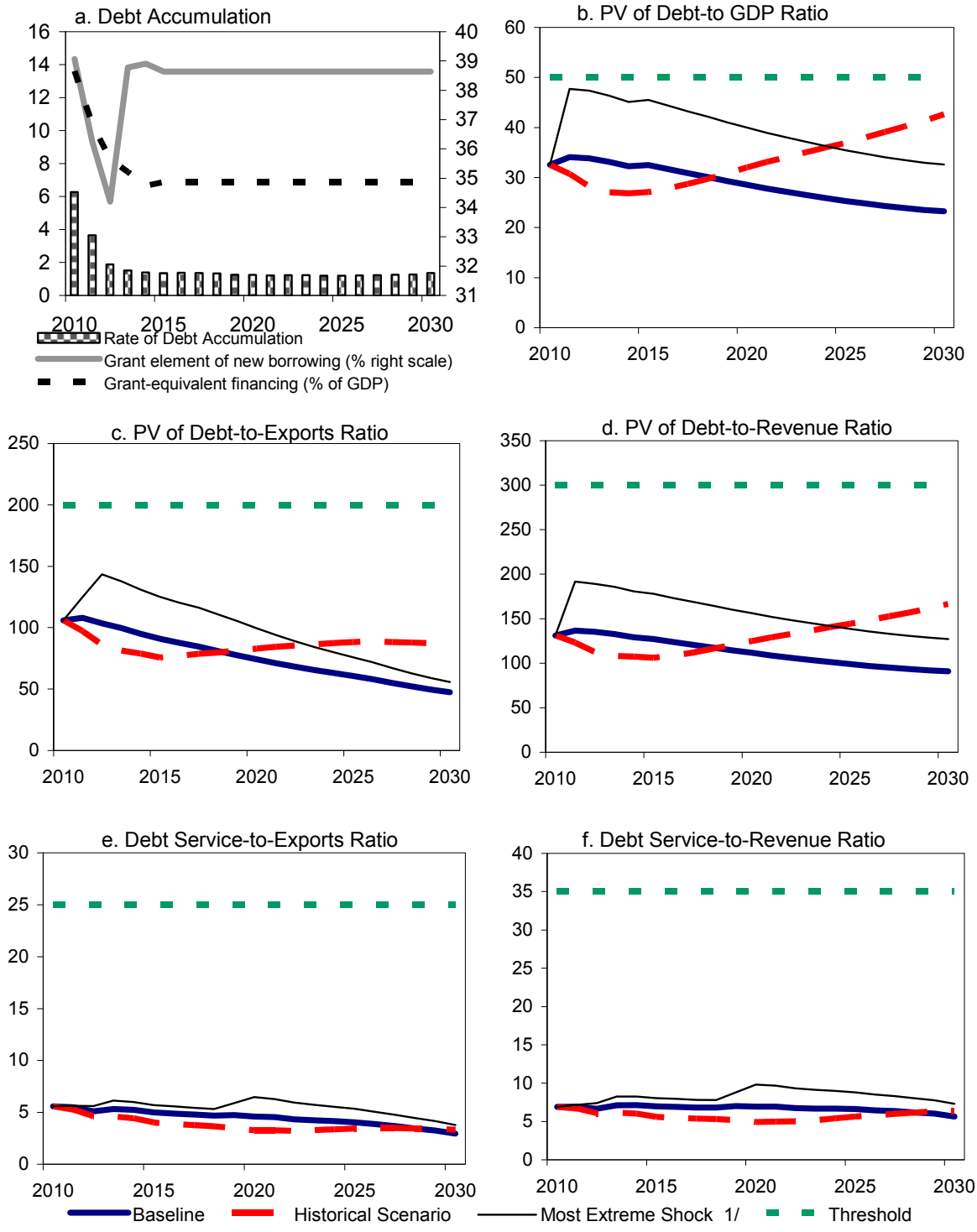
The government of Samoa's public debt management strategy is part of the annual budget submitted to parliament. The key quantitative targets include: (1) avoiding a fiscal deficit in excess of 3.5 percent of GDP, and (2) keeping public net debt at less then 40 percent of GDP. Moreover, the government is committed to obtaining concessional borrowing with a grant element of at least 35 percent. With these commitments, the government intends to maintain debt at prudent levels to provide a buffer against shocks and constrain the growth in debt service to maintain its capacity of improving public service delivery. The DSA provided in Article IV Consultations serves as input to regular updates and reviews of Samoa's attainment of the objectives set out in its debt management strategy.

Baseline/Pre-tsunami Debt Outlook

3. **The pre-tsunami debt outlook envisaged a deterioration of the debt indicators, both in nominal and in NPV terms through 2012.** This reflected the large fiscal stimulus package already present in the 2008/09 budget before the global recession hit Samoa, as well as the additional fiscal stimulus envisaged in the 2009/10 budget, as a response to the global crisis. The bulk of the stimulus was to be delivered through higher development expenditure.

4. **However, all debt burden indicators would remain well below sustainability thresholds in the baseline scenario and would decline in the medium and long run, as the fiscal stimulus unwinds.** Despite the large increase in the financing requirement to meet the budgeted expenditure plans, the NPV of the external debt was expected to peak at 35 percent of GDP in 2011 before moderating to 23 percent in 2030. All external debt and debt service indicators remained below the policy-dependent debt burden thresholds under the baseline scenario (Figure 1). The standard alternative scenarios suggest that that the projected debt path was particularly sensitive to key variables remaining at historical averages, mainly reflecting the large current account deficits of recent years.

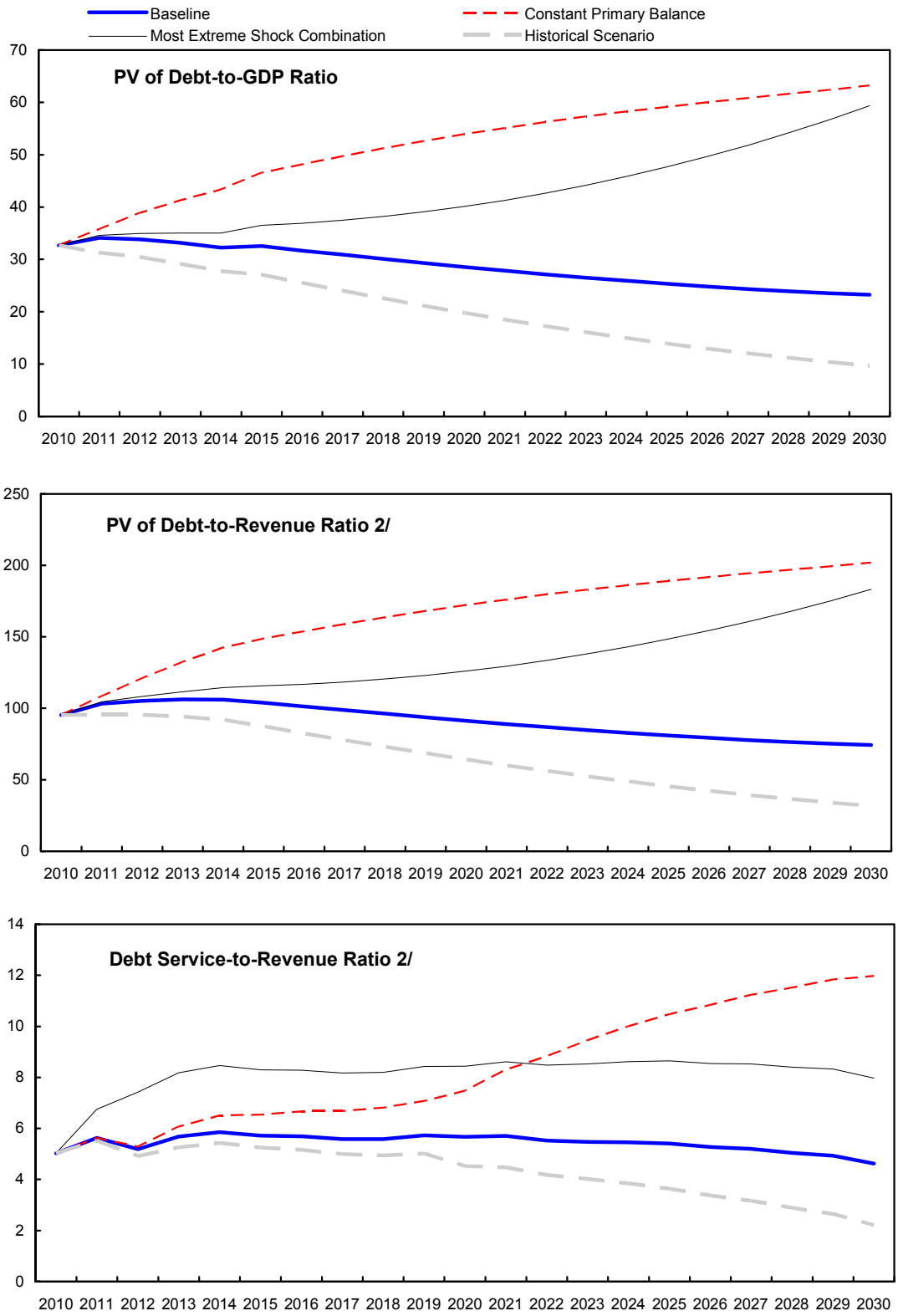
Figure 1. Samoa: Indicators of Public and Publicly Guaranteed External Debt
Under Alternatives Scenarios, 2010-2030 Pre-Tsunami Scenario 1/



Sources: Samoan authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figures b. and d. it corresponds to a one-time depreciation shock (30 percent in real terms); and in all others to a combination shock (GDP growth and the primary balance fall by 1 standard deviation from their historical average).

Figure 1a. Samoa: Indicators of Public Debt Under Alternative Scenarios (Pre-Tsunami), 2010-2030 1/



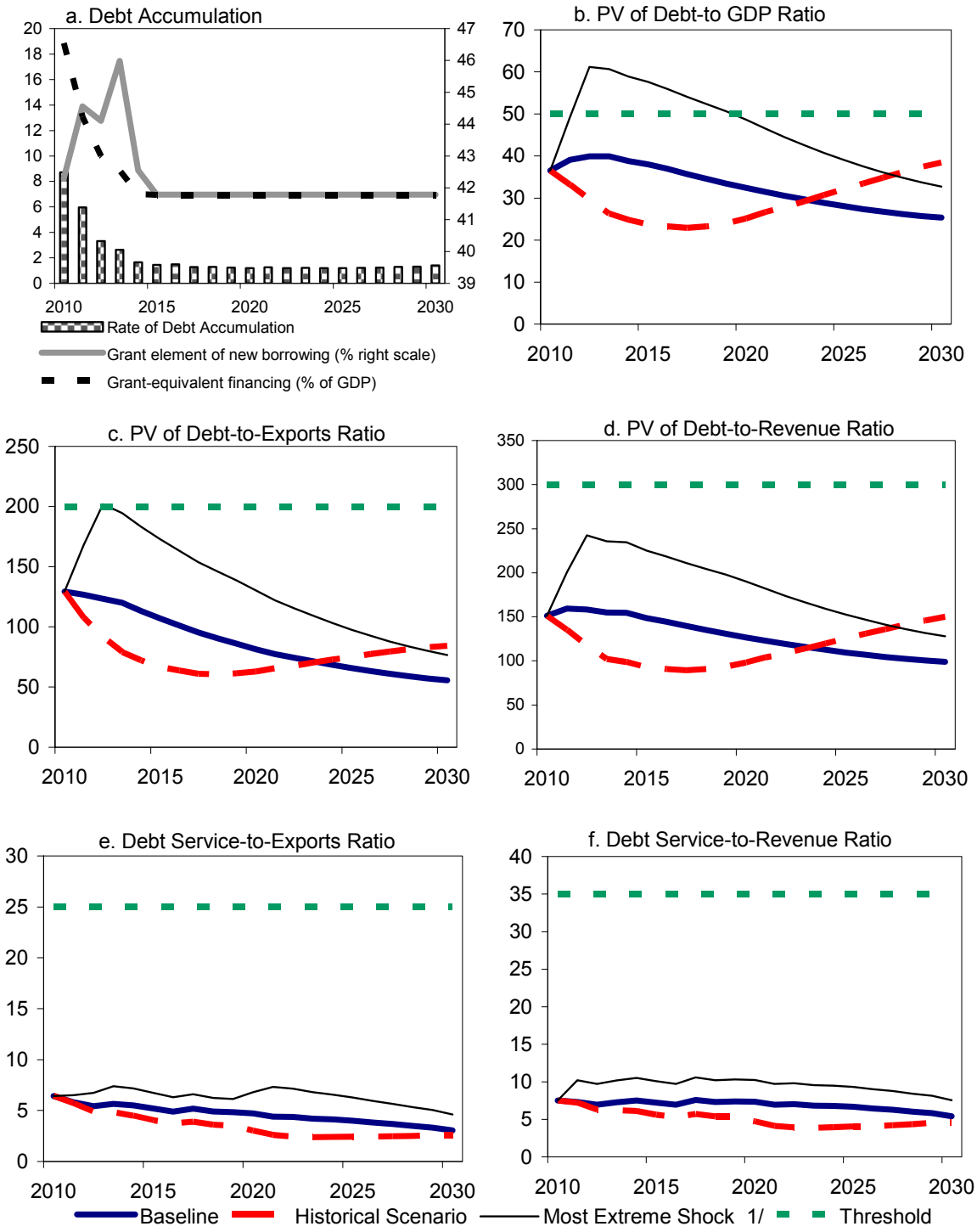
Sources: Samoan authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Post-tsunami Debt-sustainability: A Preliminary Assessment

5. **The near-term debt outlook will be adversely affected by the severe exogenous natural disaster shock Samoa has suffered.** The financing needs are assumed to be addressed by the proposed US$9 million (SDR 5.8 million) IMF *Rapid-Access Component of the Exogenous Shocks Facility (ESF-RAC*), and additional support from other international financial institutions, mainly the World Bank (up to US$40 million) and the Asian Development Bank (accelerated disbursement of US$26 million budget support), and bilateral donors (Australia, New Zealand, and the EU, about US$7 million each). To cover unidentified financing in 2009/10–2012/13, the authorities intend to approach donors to request additional support, in the form of grants and concessional loans, and also consider to temporarily lower the reserve-to-import cover to about three months. Despite the additional financing needs, bound tests suggest that the external debt would remain manageable under the shocks considered, assuming that the remaining financing gap will be filled by concessional lending and grants.

6. **Going forward, it would be important to develop a domestic bond market.** The lack of a domestic government securities market is a major shortcoming. Currently, only 0.4 percent of Samoa's public debt is domestic. Diversifying the government's low risk funding options has become increasingly important, as Samoa may graduate from less-developed country status over the medium to long term. The potentially large financing needs arising from the 2009 tsunami further underscore the benefits of developing a domestic treasury-bill and bond market. Given the support by commercial banks and the National Provident Fund to develop a domestic securities market, the government set up a working group earlier this year. Including private sector financial market participants, the objective is to reach agreement on the basic structure of a treasury-bill program for 2009–10, and introduce a bond program over the medium term.

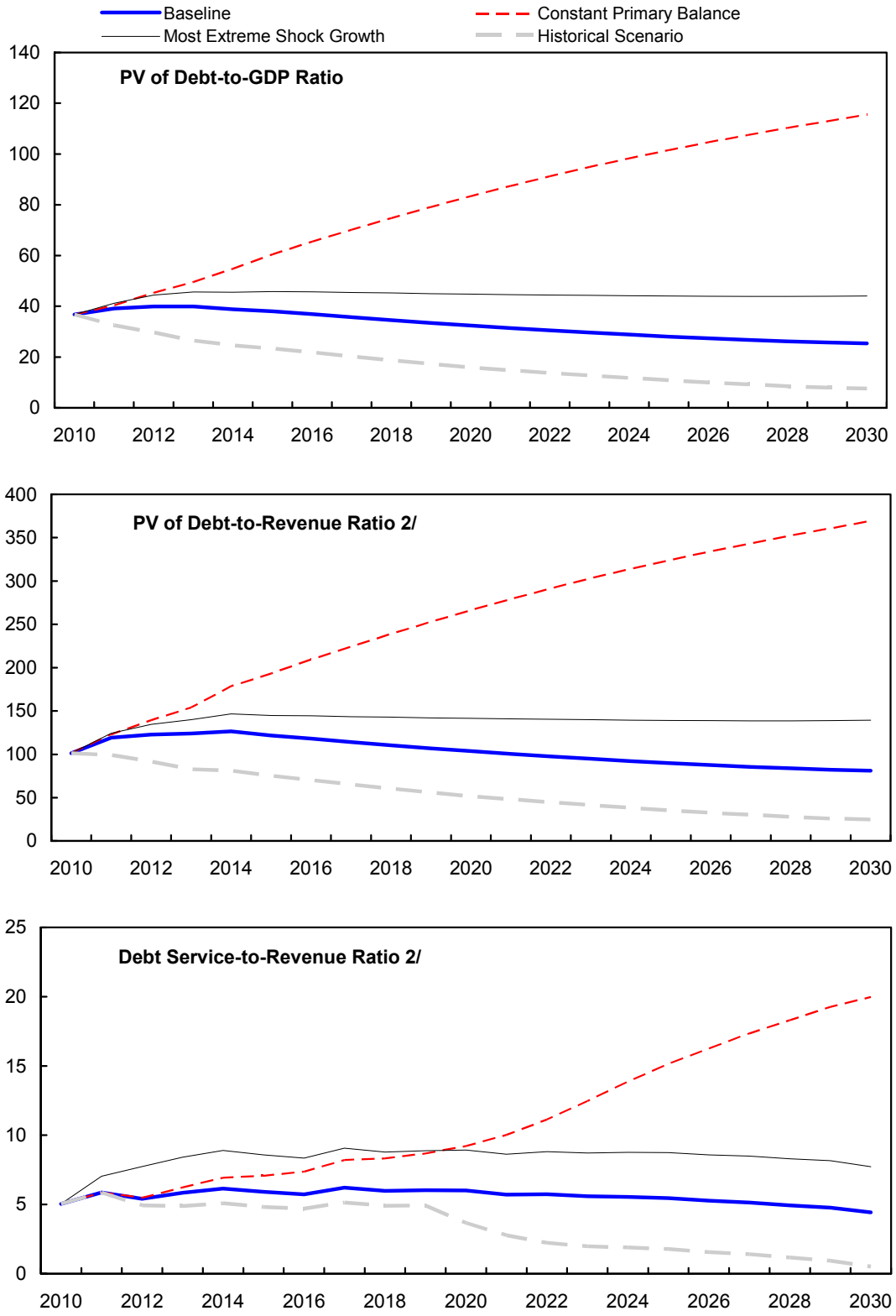
Figure 2. Samoa: Indicators of Public and Publicly Guaranteed External Debt
Under Alternatives Scenarios, Post-Tsunami 2010-2030 1/



Sources: Samoan authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figures b. and d. it corresponds to a one-time depreciation shock (30 percent in real terms); and in all others to a combination shock (GDP growth and the primary balance fall by 1 standard deviation from their historical average).

Figure 2a. Samoa: Indicators of Public Debt Under Alternative Scenarios, Post-Tsunami 2010-30 1/



Sources: Samoan authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1a. Samoa: Public Sector Debt Sustainability Framework, Baseline Scenario, Post-Tsunami 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Projections									
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	37.1	30.3	38.5			52.9	58.0	59.5	59.7	57.8	56.3		47.1	36.9	
o/w foreign-currency denominated	37.1	30.3	38.3			52.7	57.9	59.4	59.7	57.8	56.3		47.1	36.9	
Change in public sector debt	-3.0	-6.8	8.2			14.4	5.1	1.5	0.3	-1.9	-1.5		-1.6	-0.5	
Identified debt-creating flows	-6.2	-1.2	3.9			11.3	5.6	2.5	2.0	-1.4	-0.9		-1.0	-0.2	
Primary deficit	-1.4	1.6	3.4	0.6	1.8	11.1	9.3	5.3	5.1	2.0	1.7	5.8	1.8	1.9	1.9
Revenue and grants	36.4	30.9	32.2			36.3	32.9	32.6	32.1	30.7	31.3		31.3	31.3	
of which: grants	7.4	5.2	7.3			12.1	8.3	7.3	6.4	5.6	5.7		5.7	5.7	
Primary (noninterest) expenditure	35.1	32.5	35.6			47.3	42.2	37.9	37.3	32.7	33.0		33.1	33.2	
Automatic debt dynamics	-4.8	-2.8	0.6			0.2	-3.7	-2.9	-3.1	-3.4	-2.6		-2.8	-2.1	
Contribution from interest rate/growth differential	-1.3	-2.6	1.5			1.2	-1.9	-1.7	-2.0	-2.3	-2.2		-1.9	-2.0	
of which: contribution from average real interest rate	-0.4	-0.8	-0.3			0.0	-0.4	-0.5	-0.5	-0.5	-0.5		-0.5	-0.4	
of which: contribution from real GDP growth	-0.9	-1.8	1.8			1.2	-1.5	-1.3	-1.5	-1.7	-1.7		-1.4	-1.7	
Contribution from real exchange rate depreciation	-3.6	-0.2	-0.9			-0.9	-1.7	-1.1	-1.2	-1.2	-0.4		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	3.2	-5.5	4.2			3.1	-0.5	-1.0	-1.7	-0.5	-0.5		-0.7	-0.3	
Other Sustainability Indicators															
PV of public sector debt	0.0	0.0	28.3			36.7	39.1	39.9	39.9	38.8	38.0		32.4	25.4	
o/w foreign-currency denominated	0.0	0.0	28.2			36.5	39.1	39.9	39.9	38.8	38.0		32.4	25.4	
o/w external	...	...	28.2			36.5	39.1	39.9	39.9	38.8	38.0		32.4	25.4	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	0.3	2.8	4.9			12.9	11.3	7.1	7.0	3.9	3.5		3.7	3.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	0.0	0.0	87.9			101.2	119.1	122.7	124.1	126.5	121.6		103.5	81.0	
PV of public sector debt-to-revenue ratio (in percent)	0.0	0.0	113.5			151.9	159.4	158.2	155.0	154.7	148.6		126.5	99.0	
o/w external 3/	…	…	112.8			151.3	159.3	158.1	155.0	154.6	148.5		126.5	99.0	
Debt service-to-revenue and grants ratio (in percent) 4/	4.7	4.1	4.9			5.0	5.9	5.4	5.8	6.2	5.9		6.0	4.4	
Debt service-to-revenue ratio (in percent) 4/	5.8	4.9	6.4			7.6	7.9	7.0	7.3	7.5	7.2		7.3	5.4	
Primary deficit that stabilizes the debt-to-GDP ratio	1.7	8.3	-4.8			-3.3	4.2	3.9	4.9	4.0	3.2		3.4	2.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.3	5.0	-5.5	3.6	4.2	-3.0	3.0	2.2	2.6	3.0	3.0	1.8	3.0	4.6	3.1
Average nominal interest rate on forex debt (in percent)	2.2	0.7	1.7	1.7	2.2	1.6	1.1	1.1	1.0	1.0	1.0	1.1	0.9	0.9	0.9
Average real interest rate on domestic debt (in percent)	...	...	...	...	...	1.3	-2.3	2.4	2.4	2.4	3.7	1.7	3.0	...	2.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-9.2	-0.6	-2.9	-5.2	5.9	-2.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	7.0	4.9	9.8	3.9	4.6	4.1	5.6	4.0	4.0	4.0	2.7	4.0	3.9	2.3	3.8
Growth of real primary spending (deflated by GDP deflator, in percent)	0.2	0.0	0.0	0.0	0.1	0.3	-0.1	-0.1	0.0	-0.1	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	42.3	44.6	44.1	46.0	42.6	41.8	43.5	41.8	41.8	...

Sources: Samoan authorities; and staff estimates and projections.

1/ Excludes publicly guaranteed debt tentatively estimated at 4 percent of GDP.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are for the past 10 years.

Table 2a.Samoa: Sensitivity Analysis for Key Indicators of Public Debt Post-Tsunami 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	37	39	40	40	39	38	32	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	37	33	30	26	25	23	16	8
A2. Primary balance is unchanged from 2010	37	40	45	50	55	61	83	116
A3. Permanently lower GDP growth 1/	37	40	41	42	42	42	44	62
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	37	41	44	46	46	46	45	44
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	37	34	33	33	32	32	27	22
B3. Combination of B1-B2 using one half standard deviation shocks	37	34	32	33	32	32	30	28
B4. One-time 30 percent real depreciation in 2011	37	52	51	50	48	47	38	28
B5. 10 percent of GDP increase in other debt-creating flows in 2011	37	46	47	47	45	45	38	28
PV of Debt-to-Revenue Ratio 2/								
Baseline	101	119	123	124	126	122	103	81
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	101	99	92	83	81	75	52	25
A2. Primary balance is unchanged from 2010	101	123	139	154	178	193	266	369
A3. Permanently lower GDP growth 1/	101	120	126	130	136	134	139	191
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	101	124	135	140	147	145	141	139
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	101	104	101	103	105	101	86	71
B3. Combination of B1-B2 using one half standard deviation shocks	101	103	98	102	105	103	94	88
B4. One-time 30 percent real depreciation in 2011	101	159	158	156	157	150	123	88
B5. 10 percent of GDP increase in other debt-creating flows in 2011	101	141	145	146	148	142	121	91
Debt Service-to-Revenue Ratio 2/								
Baseline	5	6	5	6	6	6	6	4
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	6	5	5	5	5	4	1
A2. Primary balance is unchanged from 2010	5	6	5	6	7	7	9	20
A3. Permanently lower GDP growth 1/	5	6	5	6	6	6	7	9
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	5	6	6	6	7	7	7	8
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	5	6	5	5	6	6	5	3
B3. Combination of B1-B2 using one half standard deviation shocks	5	6	5	5	6	6	5	4
B4. One-time 30 percent real depreciation in 2011	5	7	8	8	9	9	9	8
B5. 10 percent of GDP increase in other debt-creating flows in 2011	5	6	6	7	7	6	8	6

Sources: Samoan authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a. Samoa: External Debt Sustainability Framework, Baseline Scenario, Post-Tsunami 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections													
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2016	2017	2018	2019	2020	2030	2016-2030 Average
External debt (nominal) 1/	**37.1**	**30.3**	**38.3**			**52.7**	**57.9**	**59.4**	**59.7**	**57.8**	**56.3**		**54.3**	**52.3**	**50.5**	**48.7**	**47.1**	**36.9**	
o/w public and publicly guaranteed (PPG)	37.1	30.3	38.3			52.7	57.9	59.4	59.7	57.8	56.3		54.3	52.3	50.5	48.7	47.1	36.9	
Change in external debt	-3.0	-6.8	8.0			14.4	5.2	1.5	0.3	-1.9	-1.5		-2.0	-2.0	-1.8	-1.8	-1.6	-0.5	
Identified net debt-creating flows	-4.7	1.9	-8.1			9.5	7.4	4.1	3.0	-1.7	-2.6		-3.3	-3.6	-4.6	-5.3	-5.8	-7.2	
Non-interest current account deficit	**15.2**	**5.9**	**1.5**	**7.1**	**4.0**	**19.5**	**18.7**	**14.5**	**13.3**	**8.3**	**7.7**	13.7	**7.2**	**7.0**	**6.7**	**6.6**	**6.7**	**6.6**	6.8
Deficit in balance of goods and services	28.6	20.9	21.5			38.1	36.4	33.7	34.7	30.3	29.5		28.3	27.5	26.2	25.4	24.7	18.2	
Exports	32.7	32.4	31.5			28.2	30.9	32.3	33.2	34.3	35.6		36.5	37.5	38.2	39.0	39.8	45.6	
Imports	61.3	53.4	53.0			66.3	67.3	66.1	68.0	64.6	65.0		64.8	65.0	64.4	64.3	64.5	63.8	
Net current transfers (negative = inflow)	-19.9	-22.3	-22.6	-21.4	1.7	-23.7	-24.4	-25.3	-25.8	-26.4	-26.1	-25.3	-25.5	-24.9	-24.3	-23.8	-23.2	-18.5	-21.8
o/w official	-0.4	-0.4	-0.7			-1.2	-1.0	-0.9	-0.9	-0.8	-0.8		-0.7	-0.7	-0.6	-0.6	-0.5	-0.3	
Other current account flows (negative = net inflow)	6.4	7.2	2.6			5.2	6.8	6.0	4.3	4.4	4.4		4.3	4.4	4.8	5.1	5.3	6.9	
Net FDI and official capital transfers (negative = inflow)	**-15.5**	**-5.4**	**-7.4**	**-8.0**	**3.1**	**-11.7**	**-10.4**	**-9.7**	**-9.5**	**-8.9**	**-9.2**	-9.9	**-9.4**	**-9.6**	**-10.3**	**-10.9**	**-11.5**	**-12.5**	-11.7
Endogenous debt dynamics 2/	**-4.4**	**1.4**	**-2.2**			**1.7**	**-0.9**	**-0.6**	**-0.9**	**-1.1**	**-1.1**		**-1.1**	**-1.0**	**-1.0**	**-1.0**	**-0.9**	**-1.3**	
Contribution from nominal interest rate	0.8	0.3	0.5			0.6	0.5	0.6	0.6	0.6	0.5		0.5	0.5	0.5	0.4	0.4	0.3	
Contribution from real GDP growth	-0.8	-1.9	1.5			1.1	-1.5	-1.2	-1.4	-1.7	-1.6		-1.6	-1.5	-1.5	-1.4	-1.4	-1.6	
Contribution from price and exchange rate changes	-4.4	3.0	-4.2			…	…	…	…	…	…		…	…	…	…	…	…	
Residual (3-4) 3/	**1.7**	**-8.7**	**16.1**			**4.8**	**-2.1**	**-2.6**	**-2.7**	**-0.2**	**1.1**		**1.3**	**1.6**	**2.8**	**3.5**	**4.1**	**6.7**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	0.0	
PV of external debt 4/	…	…	28.2			36.5	39.1	39.9	39.9	38.8	38.0		36.9	35.7	34.6	33.5	32.4	25.4	
In percent of exports	…	…	89.6			129.4	126.7	123.4	120.0	113.2	106.9		101.2	95.2	90.5	85.9	81.3	55.6	
PV of PPG external debt	**…**	**…**	**28.2**			**36.5**	**39.1**	**39.9**	**39.9**	**38.8**	**38.0**		**36.9**	**35.7**	**34.6**	**33.5**	**32.4**	**25.4**	
In percent of exports	**…**	**…**	**89.6**			**129.4**	**126.7**	**123.4**	**120.0**	**113.2**	**106.9**		**101.2**	**95.2**	**90.5**	**85.9**	**81.3**	**55.6**	
In percent of government revenues	**…**	**…**	**112.8**			**151.3**	**159.3**	**158.1**	**155.0**	**154.6**	**148.5**		**144.2**	**139.4**	**135.0**	**130.7**	**126.5**	**99.0**	
Debt service-to-exports ratio (in percent)	**5.2**	**3.9**	**5.1**			**6.4**	**5.8**	**5.4**	**5.6**	**5.5**	**5.2**		**4.9**	**5.2**	**4.9**	**4.8**	**4.7**	**3.0**	
PPG debt service-to-exports ratio (in percent)	**5.2**	**3.9**	**5.1**			**6.4**	**5.8**	**5.4**	**5.6**	**5.5**	**5.2**		**4.9**	**5.2**	**4.9**	**4.8**	**4.7**	**3.0**	
PPG debt service-to-revenue ratio (in percent)	**5.8**	**4.9**	**6.4**			**7.5**	**7.3**	**7.0**	**7.3**	**7.5**	**7.2**		**7.0**	**7.6**	**7.3**	**7.4**	**7.3**	**5.4**	
Total gross financing need (Billions of U.S. dollars)	0.0	0.0	0.0			0.1	0.1	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	0.0	-0.1	
Non-interest current account deficit that stabilizes debt ratio	18.2	12.7	-6.5			5.2	13.5	13.0	13.0	10.2	9.2		9.1	9.0	8.5	8.4	8.3	7.1	
Key macroeconomic assumptions																			
Real GDP growth (in percent)	2.3	5.0	-5.5	3.6	4.2	-3.0	3.0	2.2	2.6	3.0	3.0	1.8	3.0	3.0	3.0	3.0	3.0	4.6	3.1
GDP deflator in US dollar terms (change in percent)	12.3	-7.6	16.0	6.6	10.9	4.1	5.6	4.0	4.0	4.0	2.7	4.0	3.9	3.9	3.9	3.9	3.9	2.3	3.8
Effective interest rate (percent) 5/	2.2	0.7	1.7	1.7	2.2	1.6	1.1	1.1	1.0	1.0	1.0	1.1	1.0	0.9	0.9	0.9	0.9	0.9	0.9
Growth of exports of G&S (US dollar terms, in percent)	12.8	-3.8	6.2	11.3	9.8	-9.4	18.8	11.3	9.6	10.4	9.8	8.4	9.8	9.9	9.1	9.1	9.4	8.4	8.8
Growth of imports of G&S (US dollar terms, in percent)	16.4	-15.6	8.7	7.8	14.4	26.4	10.2	4.4	9.7	1.8	6.5	9.9	6.7	7.2	6.1	6.8	7.3	7.0	6.9
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	42.3	44.6	44.1	46.0	42.6	41.8	43.5	41.8	41.8	41.8	41.8	41.8	41.8	41.8
Government revenues (excluding grants, in percent of GDP)	29.1	25.7	25.0			24.2	24.5	25.2	25.7	25.1	25.6		25.6	25.6	25.6	25.6	25.6	25.6	25.6
Aid flows (in Billions of US dollars) 7/	0.0	0.0	0.0			0.1	0.1	0.1	0.1	0.1	0.1		0.1	0.1	0.1	0.1	0.1	0.2	
o/w Grants	0.0	0.0	0.0			0.1	0.1	0.0	0.0	0.0	0.0		0.0	0.1	0.1	0.1	0.1	0.1	
o/w Concessional loans	0.00	0.00	0.00			0.08	0.07	0.04	0.04	0.03	0.02		0.03	0.03	0.03	0.03	0.03	0.06	
Grant-equivalent financing (in percent of GDP)	…	…	…			18.9	13.1	10.0	8.8	7.0	7.0		7.0	7.0	7.0	7.0	7.0	7.0	7.0
Grant-equivalent financing (in percent of external financing)	…	…	…			67.2	68.8	74.6	75.6	78.4	79.9		79.9	79.9	79.9	79.9	79.9	79.9	79.9
Memorandum items:																			
Nominal GDP (Billions of US dollars)	0.5	0.5	0.6			0.6	0.6	0.6	0.7	0.7	0.8		0.8	0.9	1.0	1.0	1.1	2.2	
Nominal dollar GDP growth	14.9	-3.0	9.5			1.0	8.7	6.3	6.7	7.1	5.8	5.9	7.0	7.0	7.0	7.0	7.0	7.0	7.0

Sources: Samoan authorities; and staff estimates and projections.

1/ Public external debt. Fiscal year ending in June.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are for the past 10 years.
7/ Defined as grants, concessional loans, and debt relief.

Table 3b. Samoa: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Post-Tsunami 2010-2030
(In percent)

	Projections					
	2010	2011	2012	2013	**2020**	2030
PV of Debt-to GDP Ratio						
Baseline	37	39	40	40	**32**	25
A. Alternative Scenarios						
A1. Key variables at their historical averages in 2010-2030 1/	37	33	30	26	**25**	38
A2. New public sector loans on less favorable terms in 2010-2030 2	37	42	45	46	**43**	40
B. Bound Tests						
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	37	41	43	43	**35**	27
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	37	42	47	46	**37**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	37	43	48	48	**39**	30
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	37	45	52	51	**41**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	37	49	61	61	**49**	33
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	37	55	56	56	**45**	35
PV of Debt-to-Exports Ratio						
Baseline	129	127	123	120	**81**	56
A. Alternative Scenarios						
A1. Key variables at their historical averages in 2010-2030 1/	129	108	92	79	**63**	84
A2. New public sector loans on less favorable terms in 2010-2030 2	129	137	139	139	**108**	88
B. Bound Tests						
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	129	127	123	120	**81**	56
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	129	158	185	179	**121**	77
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	129	127	123	120	**81**	56
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	129	146	160	155	**104**	63
B5. Combination of B1-B4 using one-half standard deviation shocks	129	168	202	194	**130**	77
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	129	127	123	120	**81**	56
PV of Debt-to-Revenue Ratio						
Baseline	151	159	158	155	**127**	99
A. Alternative Scenarios						
A1. Key variables at their historical averages in 2010-2030 1/	151	135	118	102	**98**	150
A2. New public sector loans on less favorable terms in 2010-2030 2	151	172	178	180	**167**	156
B. Bound Tests						
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	151	165	169	165	**135**	106
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	151	170	185	180	**146**	107
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	151	176	189	185	**151**	119
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	151	184	205	200	**162**	112
B5. Combination of B1-B4 using one-half standard deviation shocks	151	200	243	236	**190**	128
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	151	223	221	216	**177**	138

Table 3b. Samoa: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, Post-Tsunami 2010-2030 (continued)
(In percent)

	Projections					
	2010	2011	2012	2013	**2020**	2030
Debt Service-to-Exports Ratio						
Baseline	6	6	5	6	**5**	3
A. Alternative Scenarios						
A1. Key variables at their historical averages in 2010-2030 1/	6	6	5	5	**3**	3
A2. New public sector loans on less favorable terms in 2010-2030 2	6	6	6	6	**6**	5
B. Bound Tests						
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	6	6	5	6	**5**	3
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	6	7	7	8	**7**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	6	6	5	6	**5**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	6	6	6	6	**6**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	6	6	7	7	**7**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	6	6	5	6	**5**	3
Debt Service-to-Revenue Ratio						
Baseline	8	7	7	7	**7**	5
A. Alternative Scenarios						
A1. Key variables at their historical averages in 2010-2030 1/	8	7	6	6	**5**	5
A2. New public sector loans on less favorable terms in 2010-2030 2	8	7	7	8	**9**	8
B. Bound Tests						
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	8	8	7	8	**8**	6
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	8	7	7	8	**8**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	8	8	8	9	**9**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	8	7	7	8	**9**	7
B5. Combination of B1-B4 using one-half standard deviation shocks	8	8	8	9	**10**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	8	10	10	10	**10**	8
Memorandum item:						
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	42	42	42	42	**42**	42

Sources: Samoan authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.